INCH-POUND
MIL-DTL-22240G(SH)
12 March 2019
SUPERSEDING
MIL-H-22240F(SH)
22 March 1988

DETAIL SPECIFICATION

HOSES, RUBBER, PETROLEUM-BASED FUELS AND WATER SERVICES, DISCHARGE-ONLY, SMOOTH BORE, LIGHTWEIGHT BUOYANT TYPE

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers lightweight buoyant type, smooth bore, petroleum-based fuels and discharge-only water service rubber hoses. Water services include both boiler feed water and potable water.

1.2 Classification. Hoses covered by this specification are of the following sizes and types (see 6.2).

1.2.1 Hose size. The hose size (inside diameter) is identified by two digits (see table I).

TABLE I. Hose size.

Size code number	Inside diameter (inches)
25	2½
40	4
60	6
70	7

1.2.2 Hose type. Hose type is identified by a single capital letter (see table II).

TABLE II. Hose type.

Type code letter	Type	Sizes
A	Alongside fueling	40, 60, and 70
B	Astern fueling	25 and 60
C	Alongside fueling or water transfer	25

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1.3 Part or identifying number (PIN). PINs to be used for hoses acquired to this specification are created as follows:

M	**22240**	**-**	**XX**	**X**
Prefix for Military Specification	Specification Number		Size (see table I)	Type (see table II)

Example: M22240-25B

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-24356 - Couplings and Split Clamps, 2½-, 4-, 6-, and 7-Inch Segmented, Reattachable, for Refueling-at-Sea Hose

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

21 CFR 177.2600 - Rubber Articles Intended for Repeated Use

(Copies of this document are available online at www.ecfr.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

ASTM INTERNATIONAL

ASTM D297 - Standard Test Methods for Rubber Products – Chemical Analysis

ASTM D380 - Standard Test Methods for Rubber Hose

ASTM D381 - Standard Test Method for Gum Content in Fuels by Jet Evaporation

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ASTM D412 - Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers – Tension

ASTM D413 - Standard Test Methods for Rubber Property – Adhesion to Flexible Substrate

ASTM D471 - Standard Test Method for Rubber Property – Effect of Liquids

ASTM D573 - Standard Test Method for Rubber – Deterioration in an Air Oven

ASTM D2240 - Standard Test Method for Rubber Property – Durometer Hardness

(Copies of these documents are available online at www.astm.org.)

THE ASSOCIATION FOR HOSE AND ACCESSORIES DISTRIBUTION (NAHAD)

NAHAD Handbook for the Design and Specification of Hose Assemblies

(Copies of this document are available online at https://www.nahad.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Construction – types A, B, and C. The hose shall have a rubber tube; reinforcement consisting of either braided synthetic fiber yarn (type C only) or spiral plies of synthetic fiber cord (four plies minimum for type A) fully bonded together with rubber, skim, or friction impregnated through the cord so as to create a rubber-to-rubber bond; and a rubber cover. All layers shall be bonded to their adjacent layers so as to provide a continuous rubber bond from the inner tube through the reinforcement to the outer cover. Vulcanization shall be uniform over the length of the hose. Ends of the hose shall be cut square and shall not be capped. Synthetic fiber cord shall be polyamide or polyester. Types A, B, and C shall be compatible with couplings in accordance with MIL-DTL-24356. Additionally, type B hose shall be reinforced longitudinally in such a manner to meet the requirements of 3.9.1 when tested in accordance with 4.7.9.1.

3.3 Appearance. The tube, reinforcement, and cover shall have no visual defects when the inside surface, outside surface, and ends of the hose are examined in accordance with 4.6.

3.4 Length. Unless otherwise specified (see 6.2), type A and type C hose shall be 35 feet long ±1 percent and type B hose shall be 50 feet long ±1 percent when measured as specified in 4.7.1.

3.5 Inside diameter, wall thickness, unit weight, and tube and cover thicknesses. These features of the hose, when measured in accordance with 4.7.2 through 4.7.5 respectively, shall conform to the requirements listed in table III.

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TABLE III. Dimensions, unit weight, and test pressures.

	2½-inch size [1/]	4-inch size	6-inch size [2/]	7-inch size
Inside diameter, inches	2.500±0.063	4.000 +0.062 -0.000	6.000 +0.125 -0.000	7.000 +0.125 -0.000
Wall thickness, inch	0.312±0.047	0.328±0.047	0.422±0.047	0.438 +0.047 -0.024
Tube thickness, inch	0.063 minimum	0.094 minimum	0.094 minimum	0.110 minimum
Cover thickness, inch	0.063 minimum	0.078 minimum	0.078 minimum	0.094 minimum
Unit weight, lb/ft [3/]	2.0 maximum	3.0 maximum	6.0 maximum	7.0 maximum
Proof pressure, lb/in^2 [4/]	300	300	300	300
Burst pressure, lb/in^2 [4/]	600	600 minimum	600 minimum	600 minimum

NOTES:

1/ Type B, 2½-inch size only, may vary from this table as follows: the wall thickness may be 0.353±0.047 inch in not more than two locations 180 degrees apart.

2/ Type B, 6-inch size only, may vary from this table as follows:

a. Weight: 6.3 pounds per foot (lb/ft) maximum.

b. Wall thickness: 0.475±0.047 inch, not more than two locations 180 degrees apart.

3/ Pounds per foot (lb/ft).

4/ Pounds per square inch (lb/in^2).

3.6 Proof pressure. The hose shall withstand the proof pressure specified in table III without leakage or other indication of weakness when tested in accordance with 4.7.6.

3.7 Dimensional stability. When tested in accordance with 4.7.7, the hose shall not increase in length more than 10 percent or increase in outside diameter more than 18 percent, nor shall the twist be more than 0.5 degree per foot while under the proof pressure, and shall not retain more than 2 percent change in length or 3 percent change in outside diameter after releasing the pressure. All calculations shall be based on the original dimensions of the hose.

3.8 Burst pressure. The hose shall withstand the burst pressure specified in table III without bursting, leaking, or loosening or loss of the coupling when tested in accordance with 4.7.8.

3.9 Tensile pull. When tested in accordance with 4.7.9, the hose shall withstand the tensile pull listed in table IV without any apparent damage to the hose or weakening of the coupling grips.

TABLE IV. Tensile pull for hoses.

Size (inches)	Tensile pull (pounds)
2½ (type B)	7,500
2½ (type C)	5,000
4 (type A)	5,000
6 (type A)	12,000
7 (type A)	12,000
6 (type B)	16,000

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3.9.1 Elongation and neckdown (type B). When tested in accordance with 4.7.9.1, type B hose shall have a maximum elongation of 7½ percent and a maximum neckdown of 4½ percent.

3.10 Rubber components. The properties of the rubber in the tube, cover, and friction of the hose shall conform to the requirements listed in table V.

TABLE V. Requirements for rubber components in the hose.

Rubber properties		Tube	Cover	Friction	Test paragraph
Initial:	Tensile strength, lb/in^2, minimum	2,300	1,600	Record	4.7.11.1
	Ultimate elongation, percent, minimum	300	350	Record	4.7.11.1
	Adhesion, lb/in [1/], minimum	30	30	30	4.7.11.5
After immersion in test fluid:	Tensile strength, lb/in^2, minimum	1,400	800	Record	4.7.11.2
	Ultimate elongation, percent, minimum	200	200	Record	4.7.11.2
	Volume increase, percent, maximum (no shrinkage permitted)	30	30	30	4.7.11.4
	Adhesion, lb/in, minimum	20	20	20	4.7.11.6
After oven aging:	Tensile strength, lb/in^2, minimum	2,000	1,450	-	4.7.11.3
	Ultimate elongation, percent, minimum	250	250	-	4.7.11.3
	Nonvolatile extract from tube, percent maximum	3.0	-	-	4.7.11.7
	Phosphate plasticizer in tube, qualitative determination	Absent	-	-	4.7.11.8
	Resistance of cover to ozone	-	No cracks	-	4.7.11.9
	Specific gravity	Record	Record	Record	4.7.11.11
	Durometer hardness	Record	Record	Record	4.7.11.12

NOTE:

[1/] Pounds per inch (lb/in).

3.10.1 Tube and friction. The rubber used in the tube and friction shall be a copolymer of butadiene and acrylonitrile. The tube shall be smooth, free of seams with any significant depth profile, and shall be of uniform thickness. Minor tube seams, if present, shall have rounded edges and root, shall exhibit no evidence of delamination or disbonding of tube component layers, and shall have the minimum tube thickness specified in table III. Mandrel marks or scores present in the tube shall exhibit a smooth, molded appearance and shall indicate no evidence of tearing. A few indentations arising from air entrapment between mandrel and tube may be accepted provided such minor indentation has rounded edges and root is not deeper than 10 percent of the minimum tube thickness, is firm, and has full tube thickness.

3.10.1.1 Inner tube, type C hose. The inner tube material for type C hose shall be compounded in accordance with 21 CFR 177.2600.

3.10.1.2 Water resistance, type C tube. When tested in accordance with 4.7.11.10, the extractable non-volatile matter for the type C tube shall not exceed 0.0007 ounce (21 milligrams) per square inch.

3.10.2 Cover. The rubber used in the cover shall be a blend or a copolymer of butadiene, acrylonitrile, and a polyvinylchloride.

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3.11 Branding and marking.

3.11.1 Hose. When inspected in accordance with 4.3.1.1, each length of hose shall be branded in letters at least ¼ inch high, inlaid in the cover and vulcanized thereto about 5 feet from each end. Branding shall include the following:

a. Contractor's name or trademark.

b. Month and year of manufacture.

c. Specification number and revision letter.

d. National stock number (to be furnished by the command or agency concerned).

e. Nominal hose size.

f. Hose types A, B, and C shall be branded as follows:

(1) Type A shall be marked "Alongside Fueling Only".

(2) Type B shall be marked "Astern Fueling Only".

(3) Type C shall be marked "Fuel or Water Service".

3.11.2 Striping. When inspected in accordance with 4.3.1.1, each length of hose shall be marked with a longitudinal white or red stripe along its entire length to show the natural relaxed position of the hose. This stripe shall be a white or red rubber compound securely vulcanized to the outer cover of the hose.

3.12 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. The first article shall be a first production item of the hose supplied under this specification. The first article inspection shall consist of the examination and tests specified in 4.6 and 4.7.1 through 4.7.11.

4.2.1 Rejection of first articles. Upon completion of the examination of the first article in accordance with 4.6, the presence of one or more defects shall be cause for rejection. Failure of any one test specified in 4.2 shall be cause for rejection of the first article.

4.3 Conformance inspection. Conformance inspection shall consist of the examination and tests specified in 4.3.1, 4.7.1 through 4.7.9, and 4.7.11. Sampling guidance for conformance inspection is provided in 4.4. Criteria for rejection based on conformance inspection is provided in 4.5.

4.3.1 Individual hose tests. Each hose or hose assembly shall be subjected to the following tests.

4.3.1.1 Visual inspection. The exterior surfaces, interior surfaces, and ends shall be examined for visual defects in accordance with the NAHAD Handbook for the Design and Specification of Hose Assemblies to meet the requirements specified in 3.10.1. A borescope shall be used to inspect the hose tubes. Any hose containing major or critical defects shall be rejected. Hose branding and marking shall be examined for conformance to 3.11. Any hose incorrectly marked or branded shall be rejected.

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4.3.1.2 Proof pressure. Each hose or hose assembly shall be subjected to the applicable proof pressure as specified in table III for a minimum of 2 minutes. If a hose with fittings (hose assembly) is specified (see 6.2), these fittings shall be installed prior to the proof pressure test. If the fittings are not specified, any fitting that will contain the proof pressure may be used. When temporary fittings have been installed and removed, there may be marks on the cover due to the fittings. These marks shall not be cause for hose rejection. Dimension stability tests are not required, as they are covered in the lot sampling tests. Any hose exhibiting defects due to the proof pressure shall be rejected.

4.4 Sampling for conformance inspection.

4.4.1 Sampling for examination, measurements, and tests performed on entire hose. A lot shall consist of all hoses of the same type and size, manufactured under the same conditions, and offered for delivery at one time. A random sample of hoses shall be selected from each lot in accordance with ASQ Z1.4 and shall be subjected to the examination and tests specified in 4.3.1.1, 4.7.1 through 4.7.4, 4.7.6, and 4.7.7.

4.4.2 Sampling for measurements and tests performed on sections cut from hose. Two hoses, identified as first sample hose and second sample hose, shall be selected at random from the lot (see 4.4.1). Short lengths cut from one end of these hoses shall be used for the tests specified in 4.7.5, 4.7.8, 4.7.9, and 4.7.11. The samples for the tests shall consist of the following:

a. One section not less than 32 inches in length cut from the first sample hose for the burst pressure test.

b. One section not less than 12 inches in length cut from the first sample hose for the tensile pull test.

c. One section not less than 52 inches in length cut from the second sample hose for the measurements of tube and cover thicknesses and the tests for the properties of the rubber components.

4.4.3 Sampling for tests of friction rubber properties. Six vulcanized samples of the rubber friction used to impregnate the reinforcement plies shall be used for the friction rubber tests of 4.7.11.1, 4.7.11.2, and 4.7.11.4. The samples shall be 6 by 6 inches by 0.080±0.010 inch.

4.5 Rejection based on conformance inspection.

4.5.1 Rejection on basis of wall thickness. The entire lot shall be rejected if the wall thickness of any sample hose (see 4.4.1) is less than the minimum tolerance specified in table III. The lot shall be accepted if the wall thickness of any sample hose is above the maximum tolerance, provided that:

a. The excess above the maximum tolerance is not more than 0.016 inch.

b. The excess thickness prevails over not more than 25 percent of the circumference.

c. An equal or greater proportion of the circumference has a wall thickness at least 0.016 inch under the maximum tolerance.

4.5.2 Rejection on basis of other tests performed on entire hose. The entire lot shall be rejected if any sample hose fails to meet the requirements for unit weight, proof pressure, or dimensional stability.

4.5.3 Rejection on basis of tests performed on hose sections and rubber sample. The entire lot shall be rejected if the hose sections (see 4.4.2) fail to meet the requirements for burst pressure, tensile pull, tube or cover thickness, construction, or properties of the rubber components.

4.6 Visual inspection for first article. A hose section 48 inches in length shall be cut from the center of the first article hose and cut open longitudinally to expose the tube surface. The interior, exterior, and ends of the hose section shall be examined in accordance with NAHAD Handbook for the Design and Specification of Hose Assemblies to ensure it meets the construction requirements of 3.2 and 3.10.1 and the appearance requirements of 3.3.

4.7 Tests. Unless otherwise specified herein, tests shall be performed at 80±9 °F.

4.7.1 Length. Lengths of sample hoses shall be measured in accordance with ASTM D380.

4.7.2 Inside diameter. Inside diameters of sample hoses shall be measured in accordance with ASTM D380 using plug gauges.

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4.7.3 Wall thickness. Wall thicknesses of sample hoses shall be measured at both ends of the hose using a vernier caliper. Measurements shall be made at least every 90 degrees around the circumference.

4.7.4 Unit weight. The unit weights of sample hoses (see 4.4.1) shall be determined by weighing the uncut hoses. The readings shall be accurate to the closest pound.

4.7.5 Tube and cover thicknesses. The thicknesses of the tube and cover shall be measured on specimens removed from each end of the hose section obtained in accordance with 4.4.2. The thicknesses shall be measured in accordance with ASTM D380. The respective values found for the tube thickness shall not be averaged, but shall be regarded as independent measurements, each of which shall meet the requirement. The respective values for the cover thicknesses shall be treated in the same manner. A minimum of three points on each layer shall be tested.

4.7.6 Proof pressure. Each of the sample hoses shall be equipped with couplings in accordance with MIL-DTL-24356 and subjected to the proof pressure specified in 3.6. Before starting the test, a 10-pound hydrostatic pressure shall be established in the hose, and the length, outside diameter, and zero index for the twist shall be measured in accordance with ASTM D380 in preparation for the dimensional stability test specified in 4.7.7, except that the hydrostatic pressure rate shall not exceed 1,000 lb/in^2 per minute. The proof pressure shall be held for 5 minutes and the hose and fittings examined for leakage or other indication of weakness. At the end of the 5-minute period and while the pressure is maintained, the length, outside diameter, and twist of the hose shall be measured for the determination of dimensional stability.

4.7.7 Dimensional stability. The effect of the proof test pressure (see 3.6) on the length, outside diameter, and twist of all sample hoses shall be determined in accordance with 4.7.6 and ASTM D380. The pressure shall then be reduced to 0.0 lb/in^2 and the hose left in this condition for 5 minutes. At the end of this period, the diameter and length of the hose shall be measured.

4.7.8 Burst pressure. The hose section obtained in accordance with 4.4.2 shall be equipped with couplings in accordance with MIL-DTL-24356. The burst pressure of this section in the straight condition shall then be tested in accordance with ASTM D380.

4.7.9 Tensile pull. The hose section obtained in accordance with 4.4.2 shall be equipped with couplings in accordance with MIL-DTL-24356. The assembly shall then be subjected to the tensile load as specified in 3.9, which shall be maintained for not less than 5 minutes. Upon release of the tensile load, the hose section shall be tested to the proof pressure of table III in accordance with 4.7.6.

4.7.9.1 Elongation and neckdown (type B). One complete 36- to 48-inch length of type B hose shall be subjected to a tensile pull of 3,000 pounds for the 2½-inch hose and 6,000 pounds for the 6-inch hose. The elongation and neckdown shall conform to 3.9.1.

4.7.10 Kink stress. A 48-inch-long hose section shall be prepared from the first article length including couplings in accordance with MIL-DTL-24356, and the following test steps shall be accomplished. Blank off the ends of the couplings, fill the hose section completely with reference fuel B of ASTM D471, and pressurize to 50 lb/in^2. Hold the 50 lb/in^2 pressure for 48±½ hours. At the end of the 48±½ hours, depressurize and drain the hose. Bend the hose over a 2-inch pipe such that the crease in the hose is located 1 foot from a coupling and secure that coupling to a vertical wall or frame so it will not move (see figure 1). At the other end of the hose section, attach and suspend a 650-pound weight for not less than 15 minutes. At the end of the 15 minutes, remove the weight, fill the hose completely with water, and apply a proof pressure of 300 lb/in^2 to the hose. The proof pressure shall be held for 5 minutes and the hose and fittings examined for leakage or other indication of weakness. At the end of the 5 minutes, depressurize and drain the hose. Remove the couplings and cut the hose section lengthways. Examine the hose for evidence of tube, interlayer adhesion, or reinforcement failure. Evidence of leakage, noticeable weakness, or failure of adhesion or reinforcement shall be cause for rejection of the first article.

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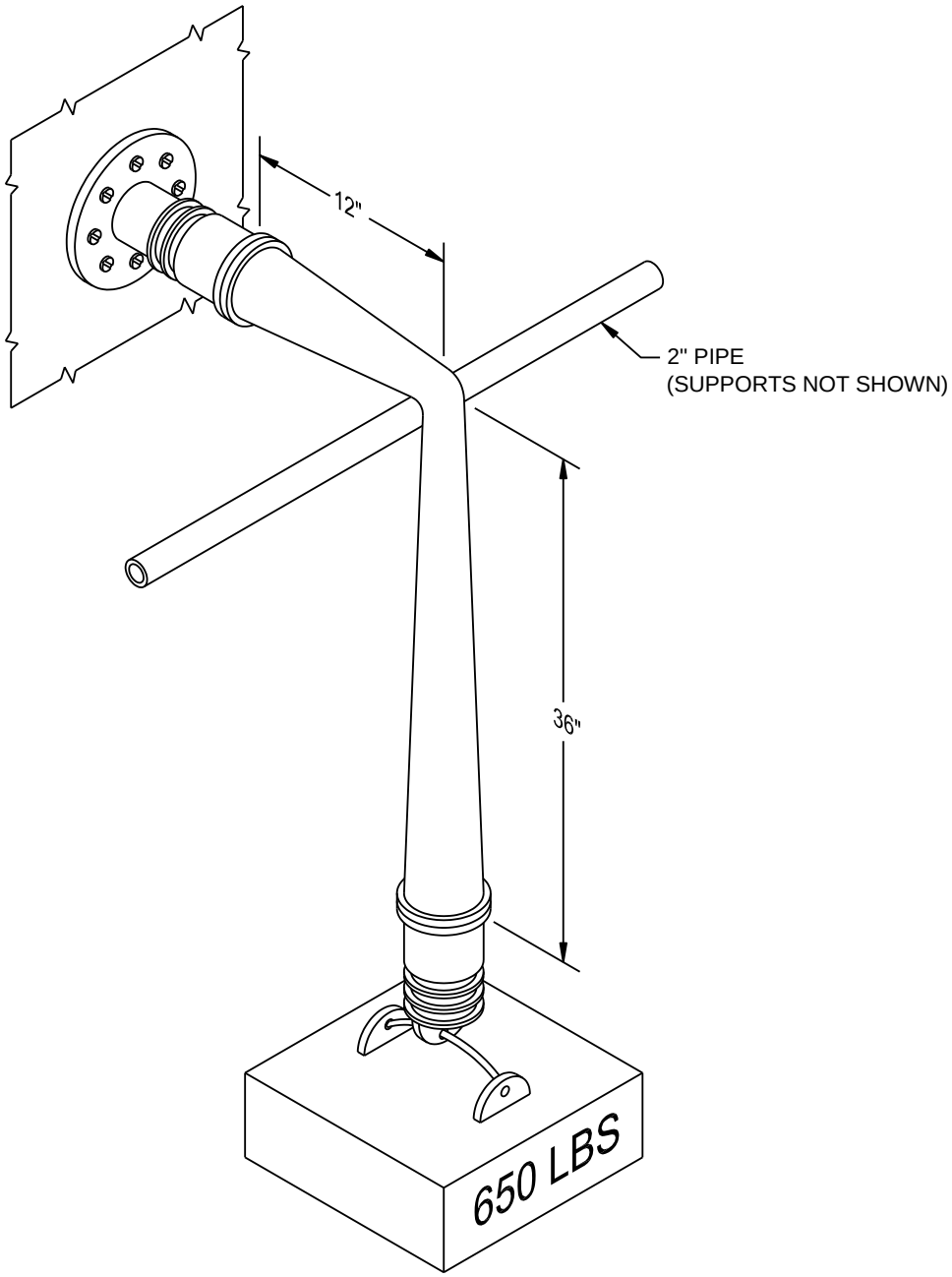


FIGURE 1. Kink stress tension test.

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4.7.11 Properties of rubber components. The properties of rubber components, except testing of the friction rubber, shall be determined on longitudinal specimens prepared from the hose section obtained in accordance with 4.4.2. After separating and buffing smooth the tube and cover samples, they shall rest not less than 30 minutes before test specimens are prepared. The testing of the friction rubber shall be determined using specimens cut from the sample obtained in accordance with 4.4.3.

4.7.11.1 Initial tensile properties. Initial tensile strength and ultimate elongation shall be in accordance with ASTM D412 using specimens cut with die C. The thickness of the die C specimen shall be 0.080±0.010 inch.

4.7.11.2 Tensile properties after immersion. The tensile strength and ultimate elongation after immersion in reference fuel B of ASTM D471 shall be determined in accordance with ASTM D471, using specimens cut with die C of ASTM D412. The thickness of the die C specimens shall be 0.080±0.010 inch. The specimens shall be immersed in fuel at a temperature of 73.4±3.6 °F for 46±¼ hours.

4.7.11.3 Tensile properties after oven aging. The tensile strength and ultimate elongation after oven aging shall be determined in accordance with ASTM D573 using specimens cut with die C of ASTM D412, except that the aging period shall be 70±¼ hours at 212±2 °F. The thickness of the die C specimen shall be 0.080±0.010 inch.

4.7.11.4 Volume change after immersion. The change in volume in percent of the original volume after immersion in reference fuel B of ASTM D471 shall be determined on specimens of the tube, cover, and friction prepared and tested in accordance with ASTM D471. The specimens shall be immersed in fuel at a temperature of 73.4±3.6 °F for 46±¼ hours.

4.7.11.5 Initial adhesion. The adhesion of the tube and cover to the reinforcement, and of the plies of reinforcement to each other, shall be determined in accordance with ASTM D413, Machine Method, and shall meet the requirements of 3.10.

4.7.11.5.1 Strip or ring specimens. Strip or ring specimens shall be prepared in accordance with ASTM D380, except that the specimen shall be a minimum of 8 inches in length for type A hose. For determining adhesion between plies of reinforcement, the plies to be separated shall be at opposing angles. Three specimens of each interface shall be tested.

4.7.11.6 Adhesion after immersion. The adhesion of the tube and cover to the reinforcement, and of the plies of reinforcement to each other, shall be determined after the test specimens (see 4.7.11.5.1) have been immersed in reference fuel B of ASTM D471 for 46±½ hours at 80±9 °F. The test procedure for adhesion shall be in accordance with ASTM D413, Machine Method, except that the wet specimen shall be tested within 15 minutes after it is removed from the fuel. The adhesion values shall meet the requirements of 3.10.

4.7.11.7 Nonvolatile extract.

4.7.11.7.1 Test specimens. Test specimens shall consist of strips about $\frac{1}{16}$ inch square by 2 inches long, totaling about 0.011 pound (5 grams) in weight, cut from buffed pieces of the tube of the hose.

4.7.11.7.2 Procedure. The test specimen shall be weighed and placed in an extraction thimble in the extracting apparatus in accordance with ASTM D297, and 3.38 ounces (100 milliliters) of reference fluid A of ASTM D471 shall be added to the flask. The specimen shall be extracted for 46±½ hours at such a rate that 2½ to 3½ minutes are required to fill and empty the extraction cup. The extract shall be poured into a weighed container. The specimen and flask shall be rinsed with 0.845 ounce (25 milliliters) of fresh reference fuel A and the wash added to the extract. The nonvolatile residue shall be determined from the extract in accordance with ASTM D381, except that the evaporation time shall be 45 minutes. The weight of the residue shall be expressed as a percent of the original weight of the specimen. The test shall be performed in duplicate and the results averaged.

4.7.11.8 Phosphate plasticizer test. Phosphate plasticizer shall be determined theoretically by compound formula review or qualitatively.

4.7.11.9 Resistance to ozone.

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4.7.11.9.1 Test specimens. Four 5- by 1-inch specimens of cover material shall be cut from the sample section of hose. The thickness of the specimens shall be 0.080±0.010 inch. Two specimens shall be cut along the longitudinal direction of the hose and the other two shall be cut in the transverse direction of the hose.

4.7.11.9.2 Apparatus. Apparatus shall consist of a test chamber with an ozone-generating source outside the chamber maintaining an ozone-air atmosphere in the chamber; a circulating air oven in accordance with ASTM D573; and two pairs of metal plates, ⅜ by 1½ by 8 inches each, with bolt holes spaced centrally ½ inch from each end, and with bolts and nuts for clamping the specimens.

4.7.11.9.3 Procedure. The specimen shall be placed between the metal plates and clamped so that the distance across one of the loops thus formed is 2 inches (see figure 2). The bulk of the specimen shall be reduced by cutting away the loop that projects from the side opposite the measured loop. The cut edges of the specimen shall be coated with paraffin to seal them from ozone. The clamped specimens shall be conditioned in an air oven for 16 to 24 hours at 100±2 °F. The specimens shall then be placed in the ozone test chamber for 166±½ hours at 105±2 °F. The concentration of ozone in the chamber shall be 1.00±0.05 parts of ozone per one million parts of air by volume. Specimen shall be examined at the end of this period for cracking of the hose cover using 7× magnification to determine conformance with the requirements specified in 3.10.

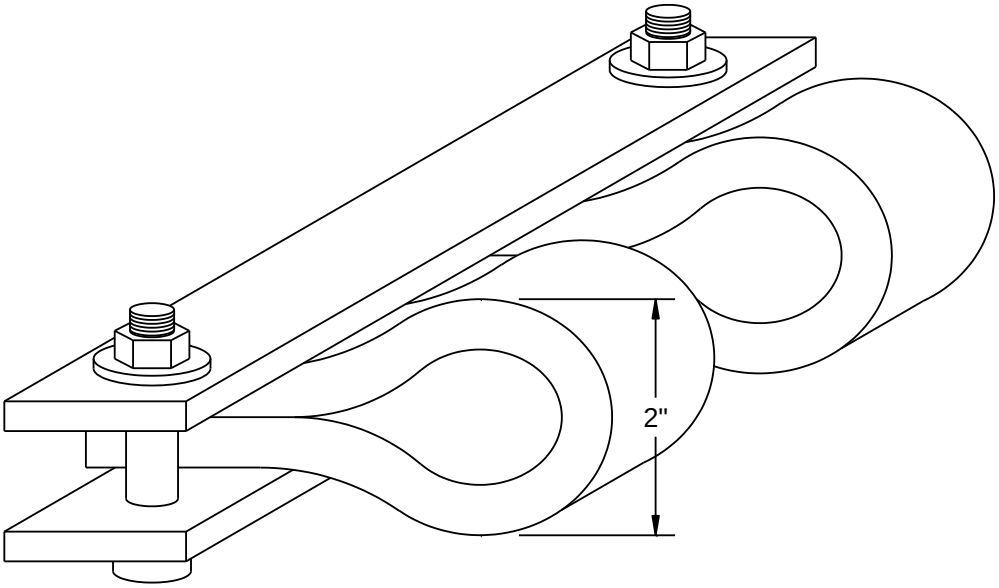


FIGURE 2. Ozone resistance test.

4.7.11.10 Resistance to water. Type C hose only.

4.7.11.10.1 Hose sampling. The inner tube surface of one hose sample per lot shall be extracted in distilled water at reflux temperature in accordance with the following procedure:

a. Remove a 1-inch ring of rubber inner tube and cut or buff away the outer surface until the surface is smooth.

b. Cut a sufficient length of inner tube to obtain approximately 0.36 ounce (10 grams), based on a minimum thickness of 3⁄64 inch.

c. Subject sample to distilled water at reflux temperature for 7 hours in an extraction apparatus in accordance with ASTM D297.

d. Filter the solution through grade 40 Whatman filter paper, or equal, and collect the extract in a tared container.

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e. Extract the rubber sample in an additional 1.7 to 2.4 ounces (50 to 75 cubic centimeters) of distilled water at reflux temperature for 2 more hours and repeat step d. Combine the extract solutions and evaporate to dryness. The container and residue shall be dried in an oven at 221±5 °F for 1 hour, cooled in a desiccator, and weighed. The weight of the residue shall be recorded. A blank shall be run using the same amount of distilled water. After making allowance for the blank, the weight of the residue shall be recorded to the nearest 0.01 ounce (milligram).

f. Record the amount of matter extracted from the rubber in milligrams per square inch of surface, determined as follows:

(1) Length (inches) × width (inches) × 2 + total surface area (square inches).

(2) Milligrams per square inch = weight of extract/total surface area.

4.7.11.11 Specific gravity. The density of tube, friction, and cover material shall be measured in accordance with the Hydrostatic Method of ASTM D297. The value shall be converted to the equivalent specific gravity and recorded.

4.7.11.12 Hardness. The durometer hardness of tube, friction, and cover material shall be measured and the values recorded in accordance with ASTM D2240.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Hose is intended for use in, but not limited to, fueling-at-sea (FAS). FAS is a U.S. Navy-unique operation in which fuel is transferred between ships while at sea and in motion. Existing stock manufactured to MIL-H-22240F, dated 22 March 1988, is acceptable for use until depleted. Future requisitions for hoses calling out MIL-H-22240 should refer to this version of the specification. The maximum operating pressure of hoses is 150 lb/in^2.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Types and sizes (see 1.2).

c. When first article is required (see 3.1).

d. Length and size of hoses required and whether or not residual lengths will be accepted (see 3.4).

e. Hose with fittings (see 4.3.1.2).

f. Packaging requirements (see 5.1).

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Subject term (key word) listing.

Hose, fuel

Hose, refueling, astern

Hose, water

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6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activity:
DLA – CC

Preparing activity:
Navy – SH
(Project 4720-2018-012)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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